Quantifying Aleatoric and Epistemic Dynamics Uncertainty via Local Conformal Calibration

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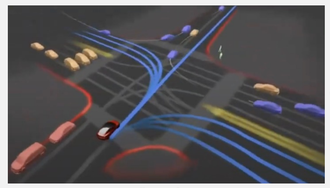
scan me for more!

1. Motivation

- *Safety-critical autonomous systems* *act under uncertainty*

- We provide probabilistically valid prediction regions for any *finite calibration dataset*, *dynamics predictors* (outputting Gaussians), *unknown* true dynamics, and *aleatoric perturbations*

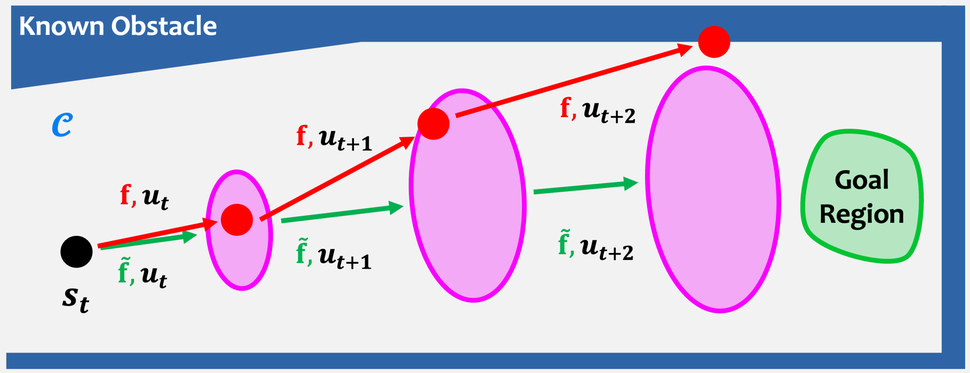
Can guarantee safety at a user-defined reliability rate $\in (0,1)$


Safe Plan

Error accumulation

Unsafe Execution

2. Problem



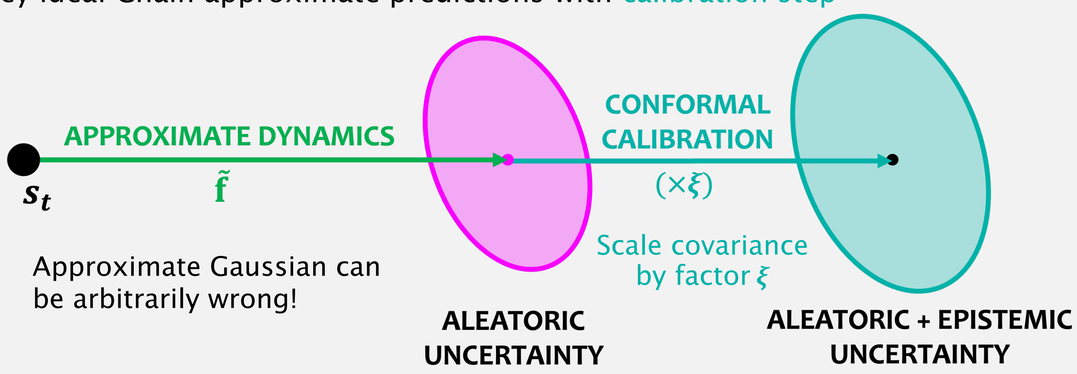
$$\min_{u_{t:t+H-1}} \quad J(s,u)$$

s: Robot State u: Control Input

$$\text{s.t.} \quad \mathbb{P}(s_{\tau+1} \in \mathcal{C}) \geq (1-\alpha), \quad \forall \tau \in \{t, \dots, t+H-1\}$$
$$s_{\tau+1} \sim \tilde{\mathbf{f}}(s_\tau, u_\tau), \quad \forall \tau \in \{t, \dots, t+H-1\}$$

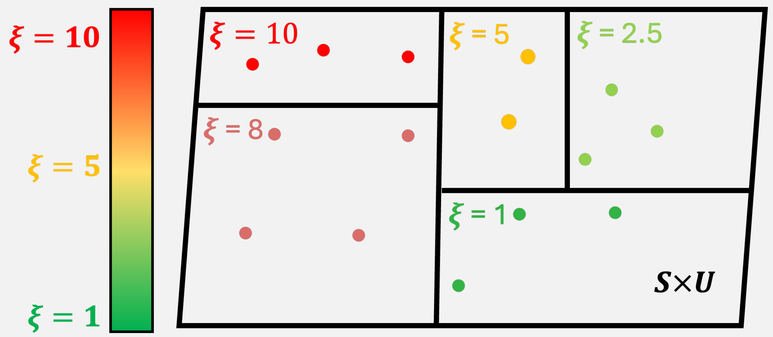
3. Method

Key idea: Chain approximate predictions with calibration step



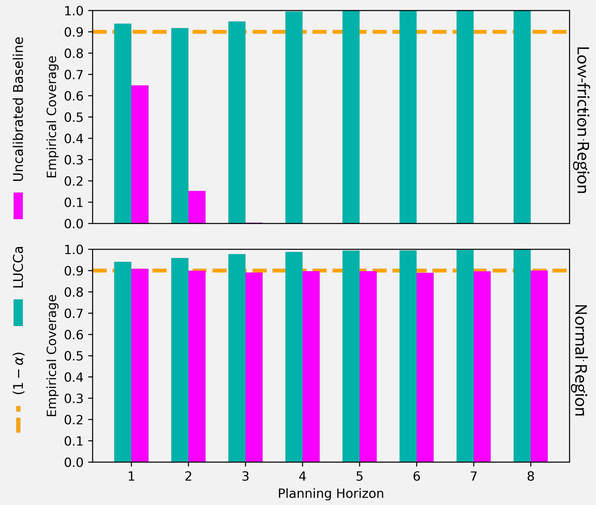
Approximate Gaussian can be arbitrarily wrong!

Model accuracy might vary along state-action space

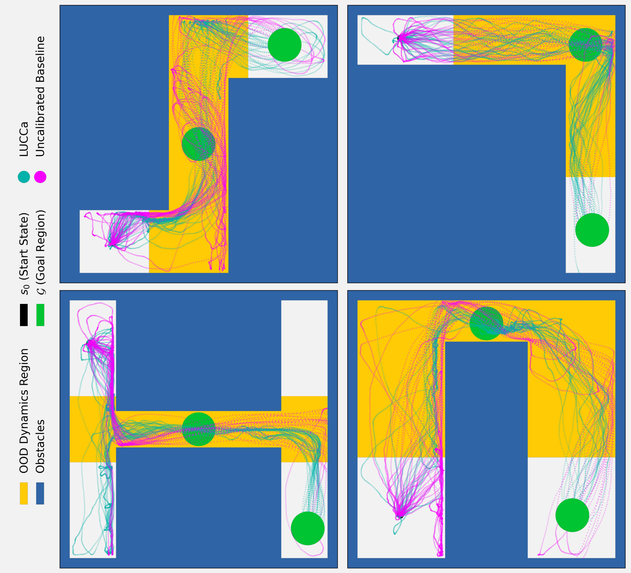


4. Results

Metrics on double integrator:
- Collision-rate ↓
- Success rate ↑
- Avg Duration ↑



Computational Cost
- Offline: full process takes 0.17 secs
- Online: overhead of 0.3 ms/planning step



5. Conclusion

- Calibrated predictors of *arbitrary performance*
- Performed *state-action dependent calibration*
- *Reduced collision rate* on all four maps to below user-set rate

f: True Dynamics $\tilde{\mathbf{f}}$**:** Approximate Dynamics J**:** Objective Function α**:** Acceptable Failure-rate H**:** Planning Horizon $\mathcal{C}$**:** Safe Region

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NSF ONR